

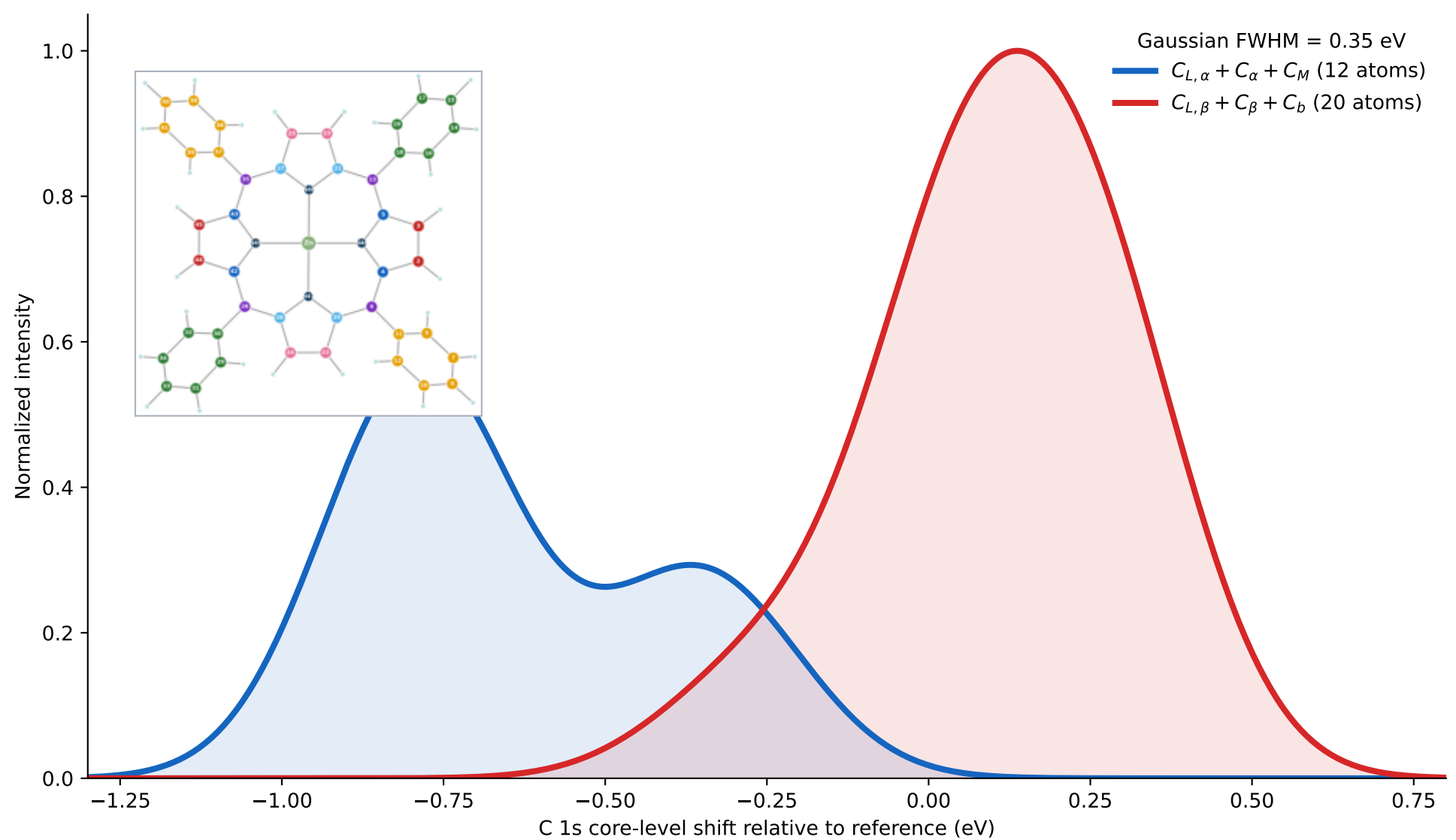
Core-Level-Shift Plot Comparison

Initial-state C 1s and N 1s spectra

Included systems

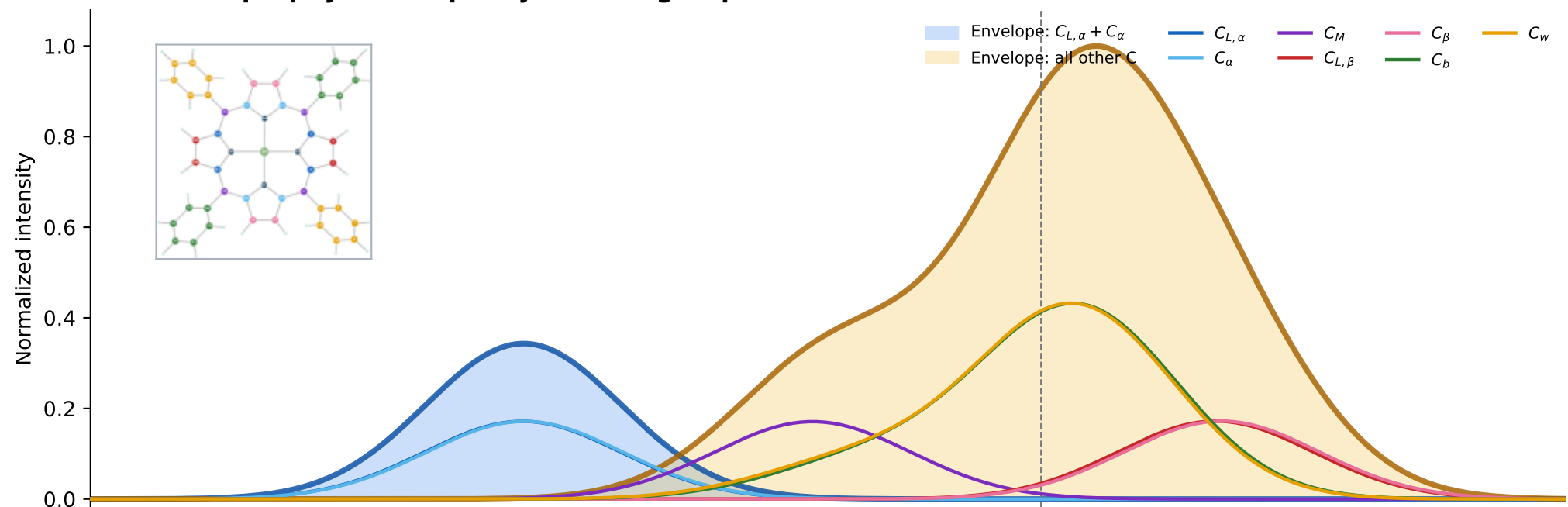
- Planar and planar-COOH comparison
- Cross
- TwistedH2 and TwistedO2
- TWco and TWDCOH
- Five-structure and focused twisted comparisons

The individual spectra are retained at their original vector quality.
PDF bookmarks provide direct navigation to every plot.

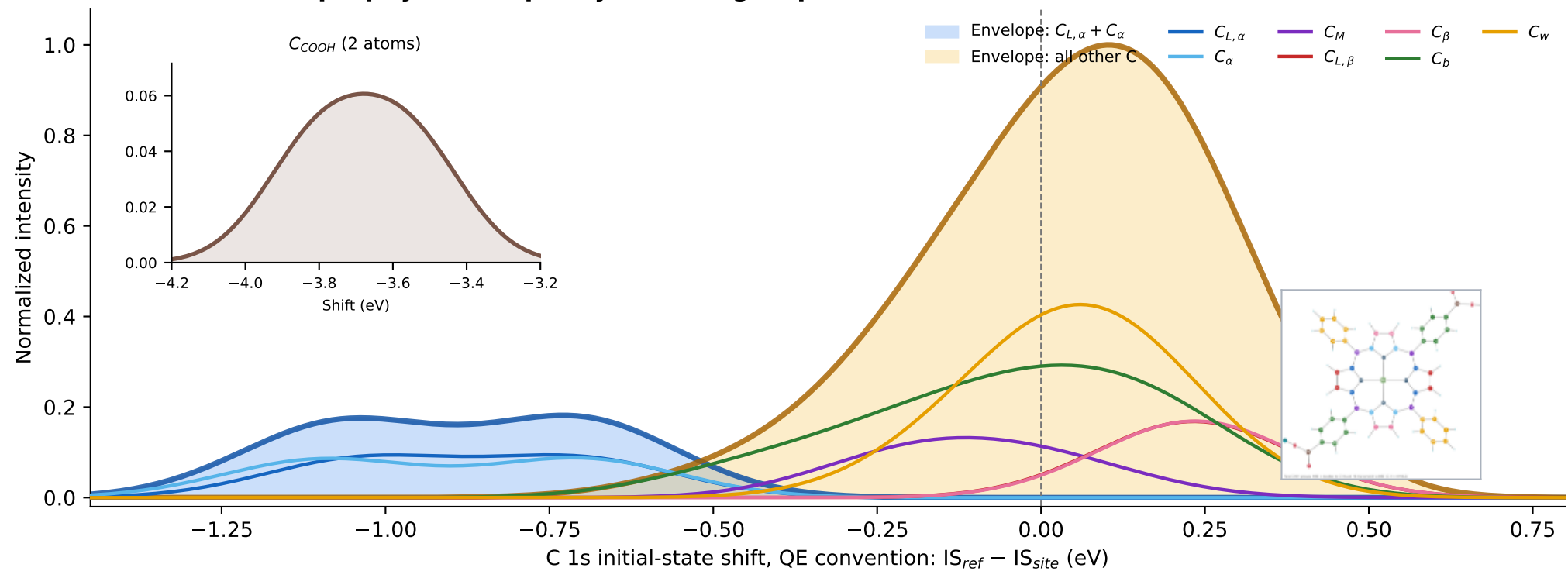


Planar and PLDC-COOH C 1s envelopes (Gaussian FWHM = 0.35 eV)

Planar: all porphyrin and phenyl carbon groups

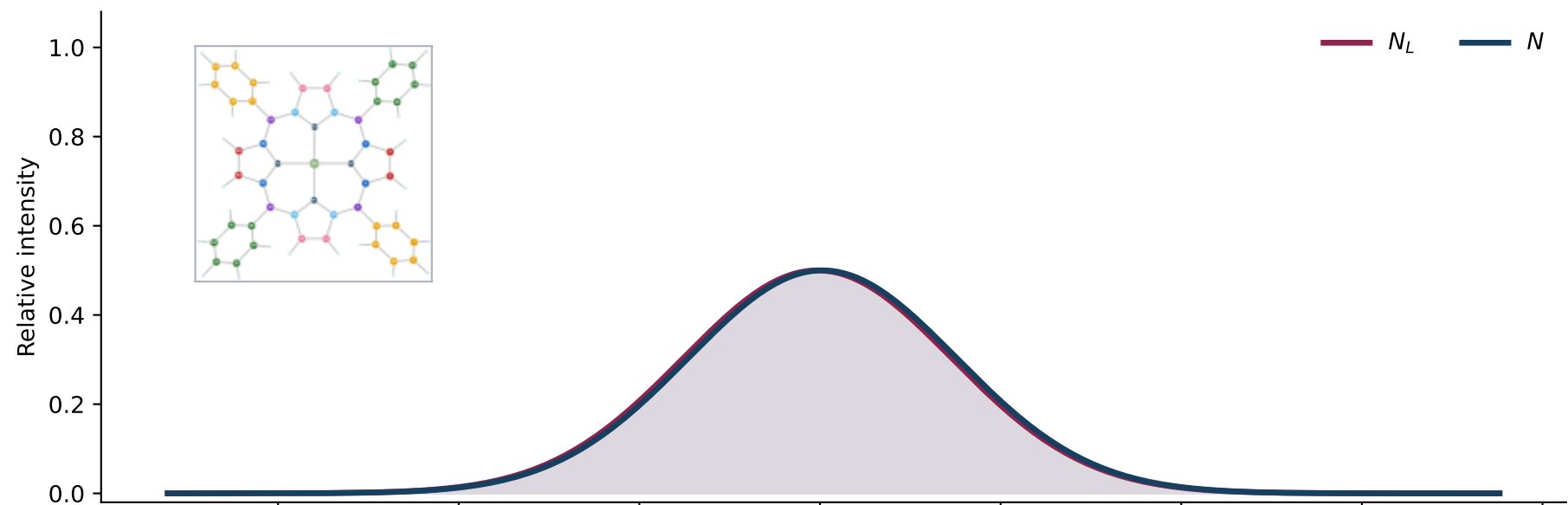


PLDC-COOH: all porphyrin and phenyl carbon groups

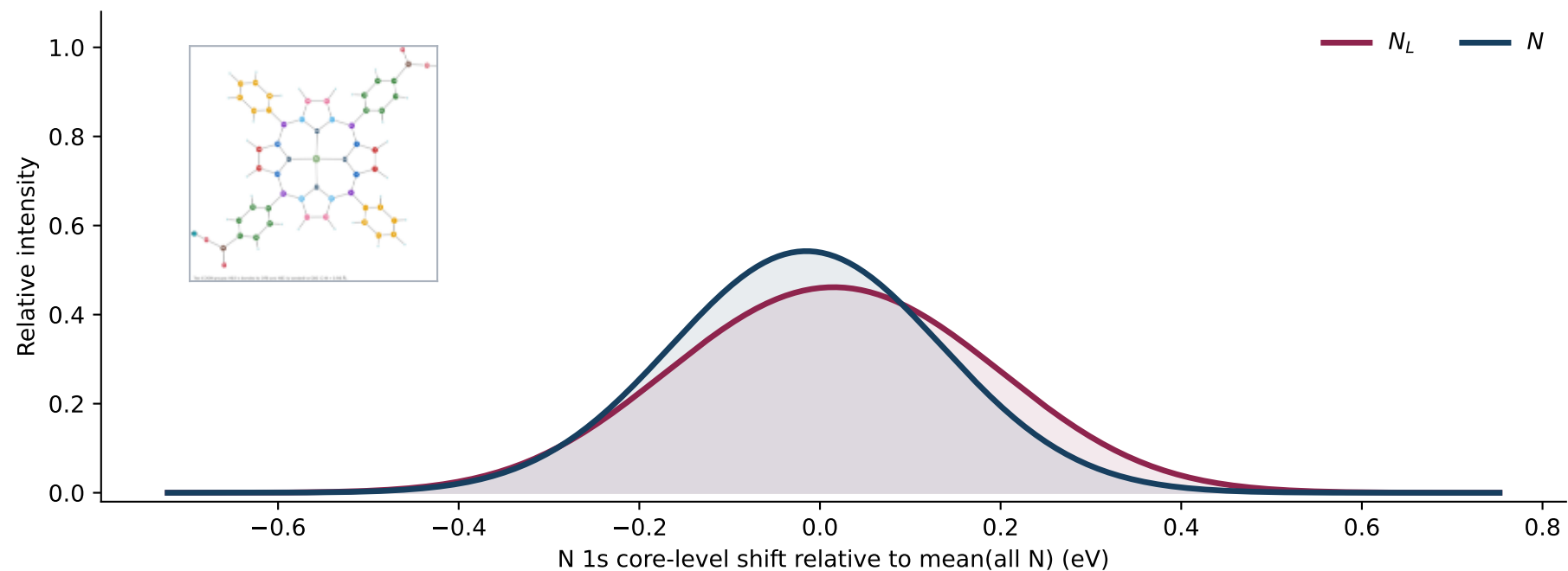


Planar and PLDC-COOH N 1s initial-state CLS (Gaussian FWHM 0.35 eV)

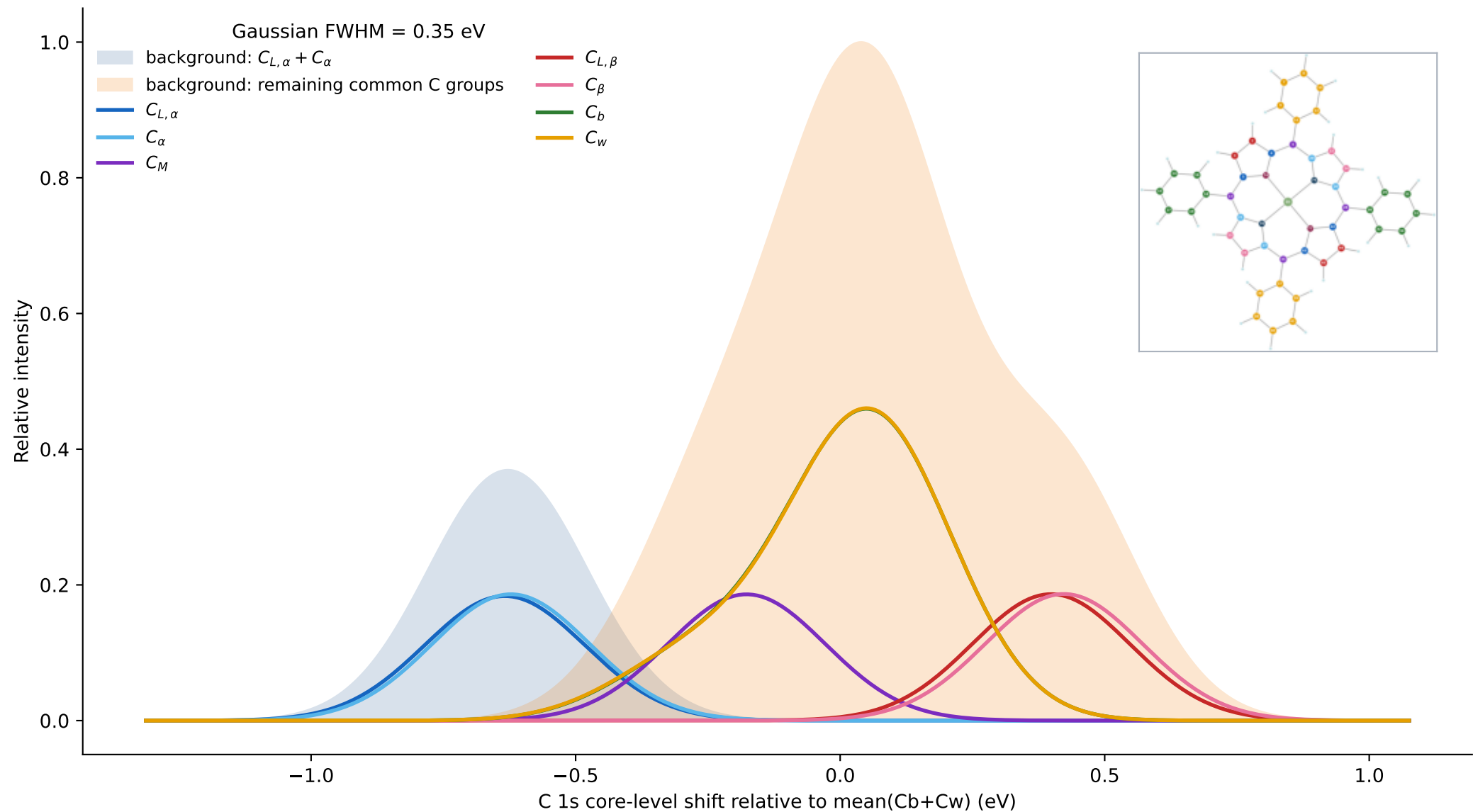
Planar



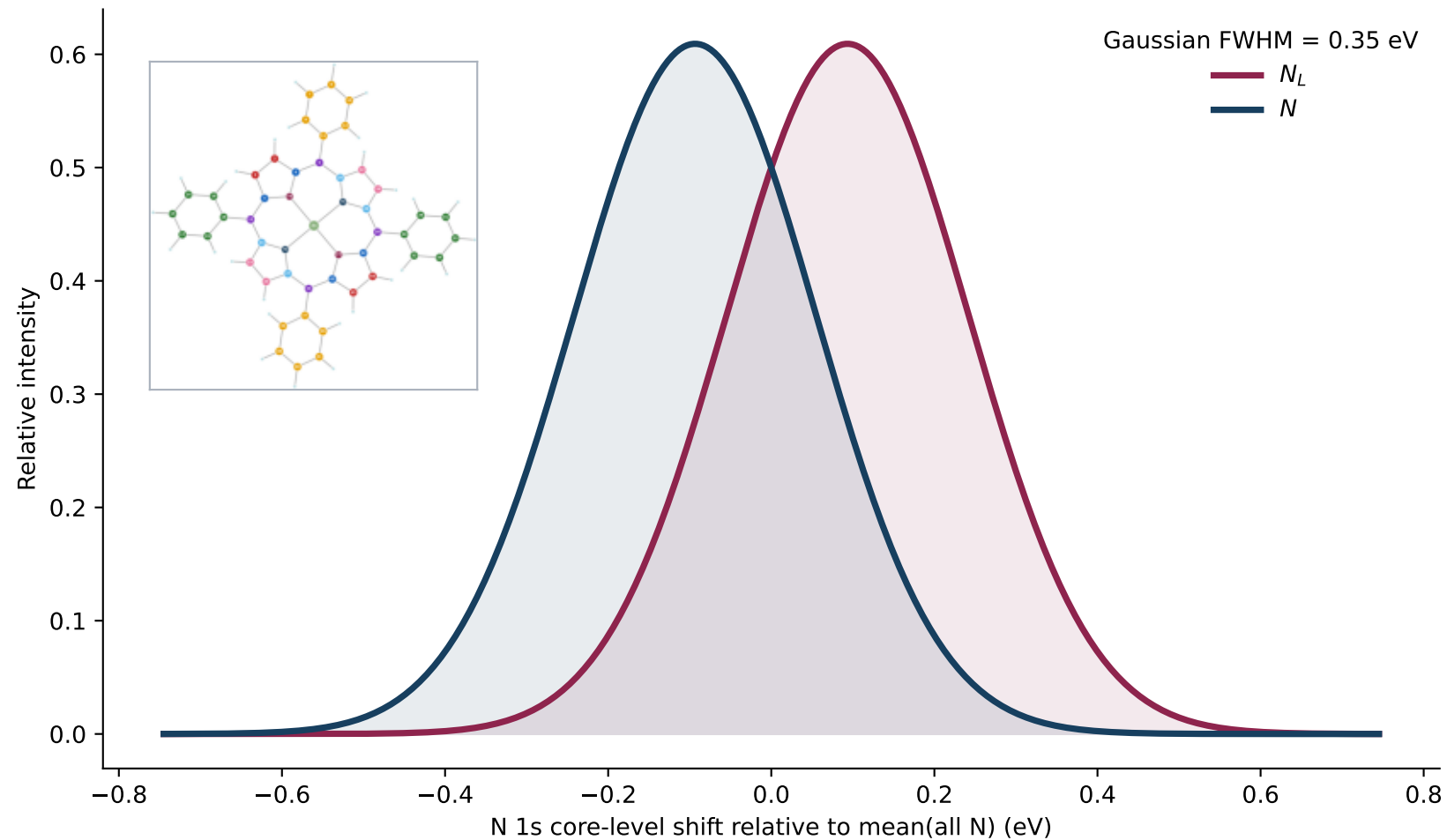
PLDC-COOH



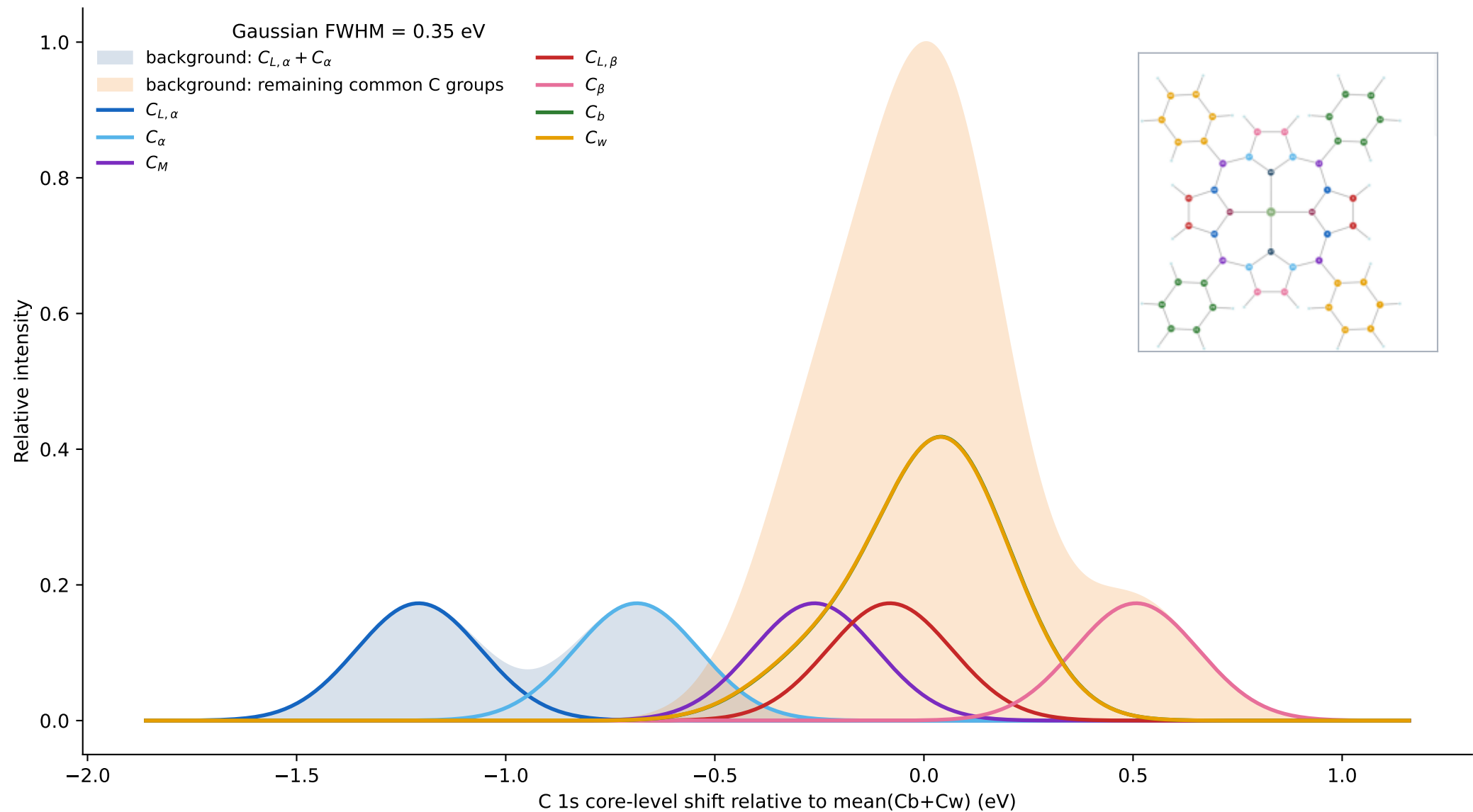
cross: C 1s initial-state core-level shifts



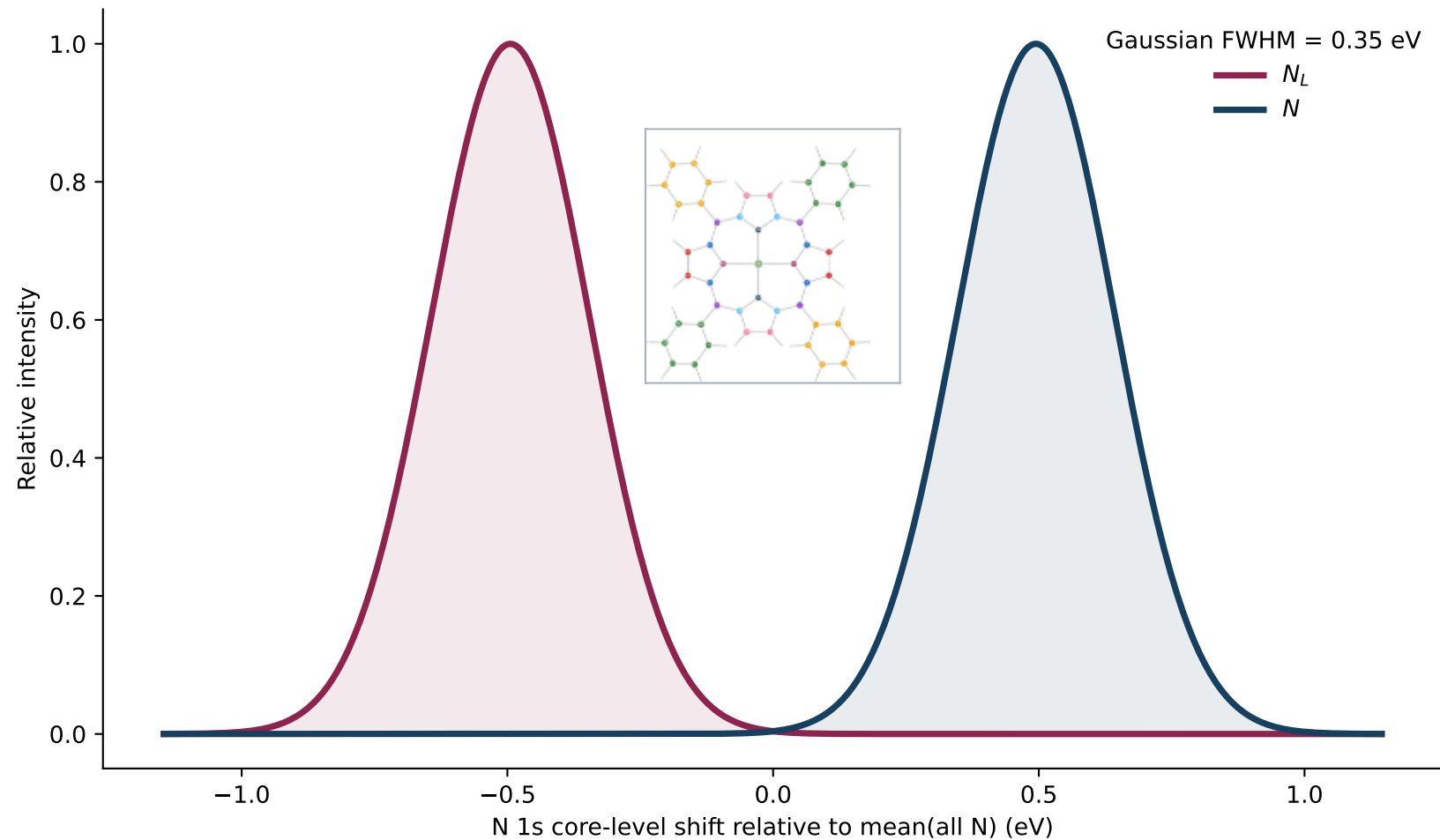
cross: N 1s initial-state core-level shifts



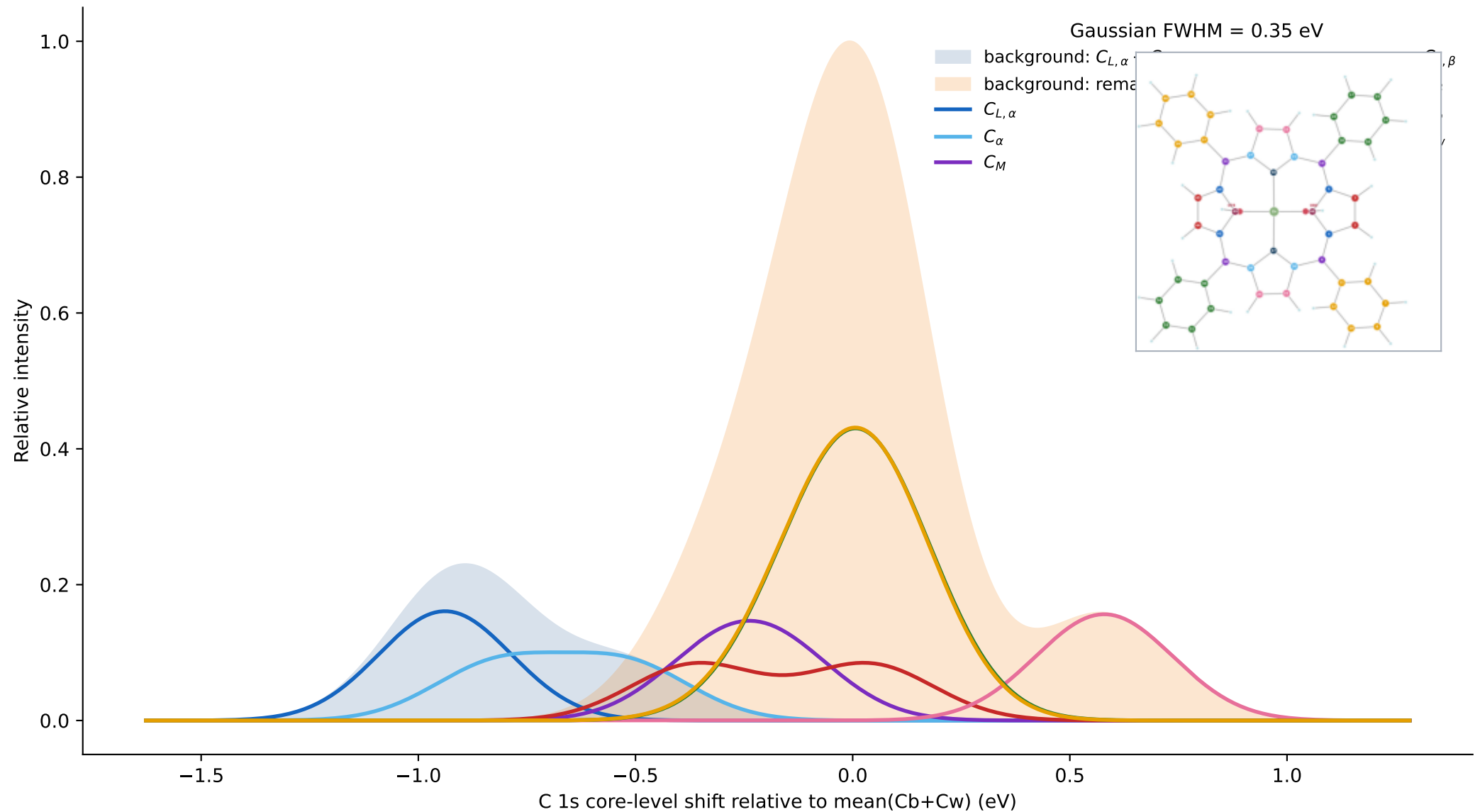
twistedH2: C 1s initial-state core-level shifts



twistedH2: N 1s initial-state core-level shifts



twistedO2: C 1s initial-state core-level shifts

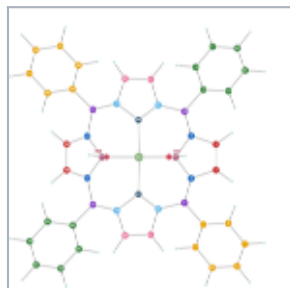


twistedO2: N 1s initial-state core-level shifts

Gaussian FWHM = 0.35 eV

N_L

N



Relative intensity

1.0
0.8
0.6
0.4
0.2
0.0

-2

-1

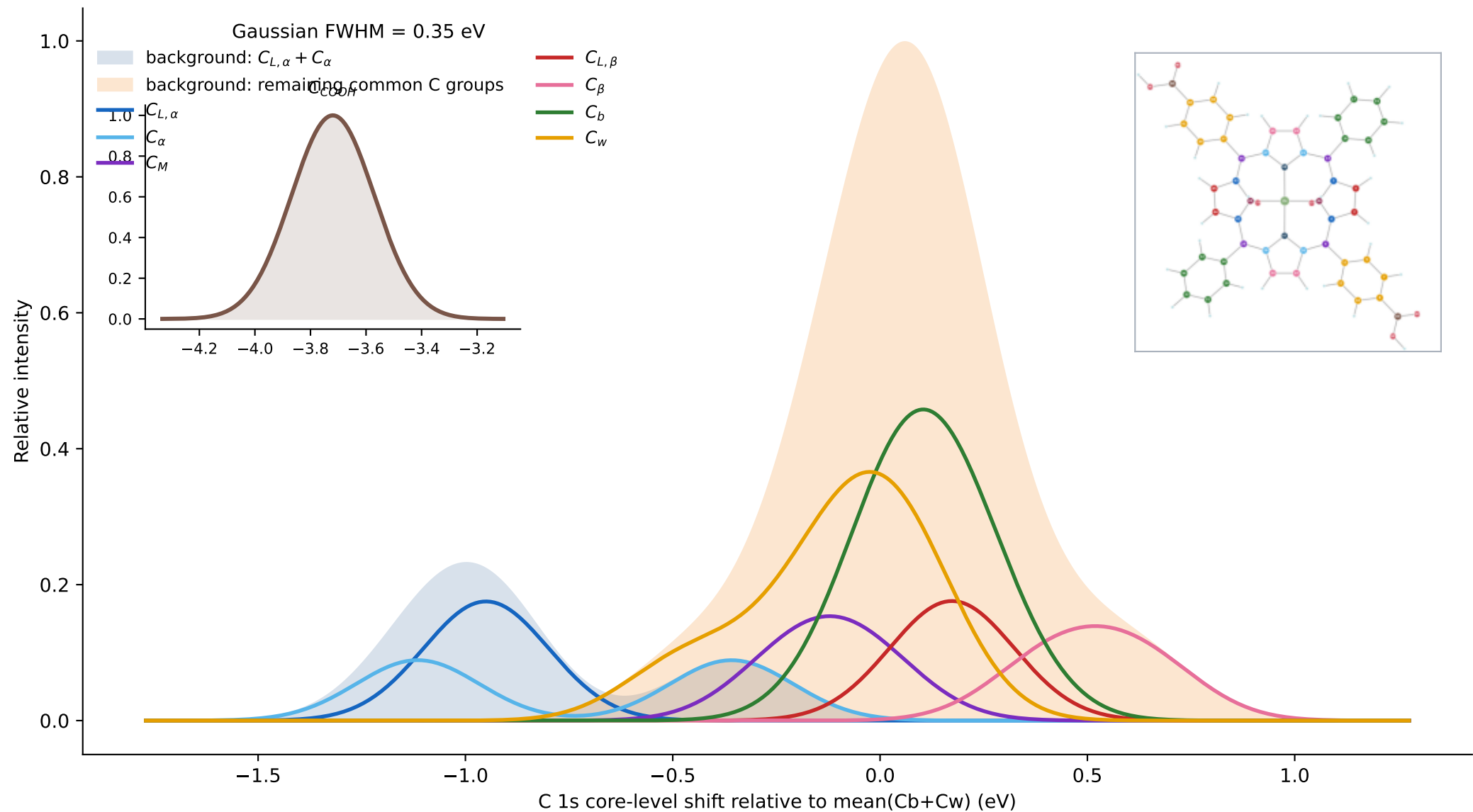
0

1

2

N 1s core-level shift relative to mean(all N) (eV)

TWco: C 1s initial-state core-level shifts

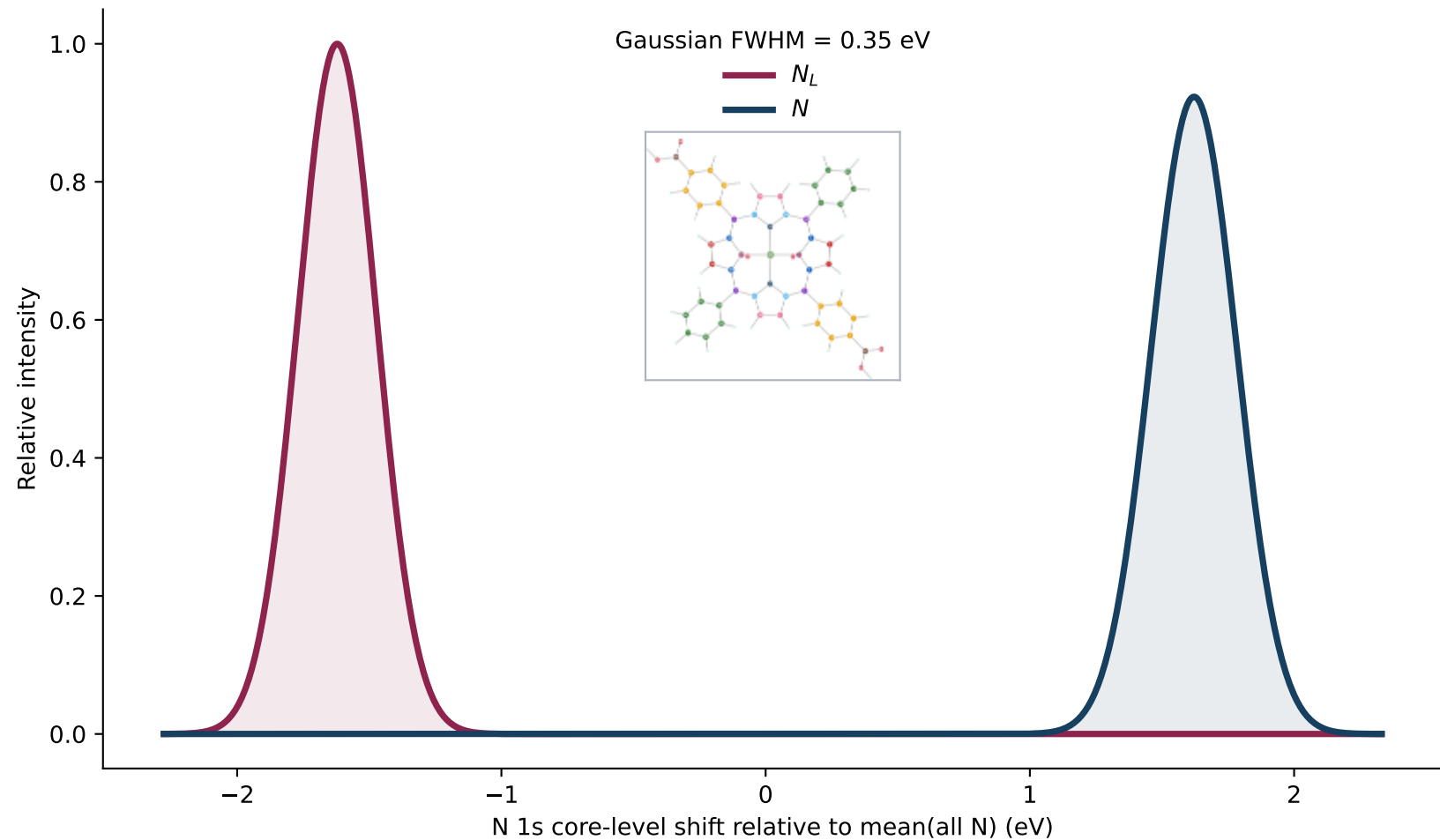
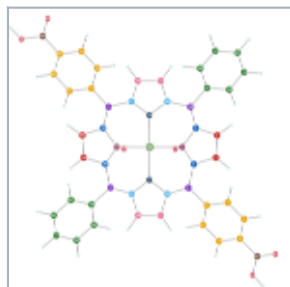


TWco: N 1s initial-state core-level shifts

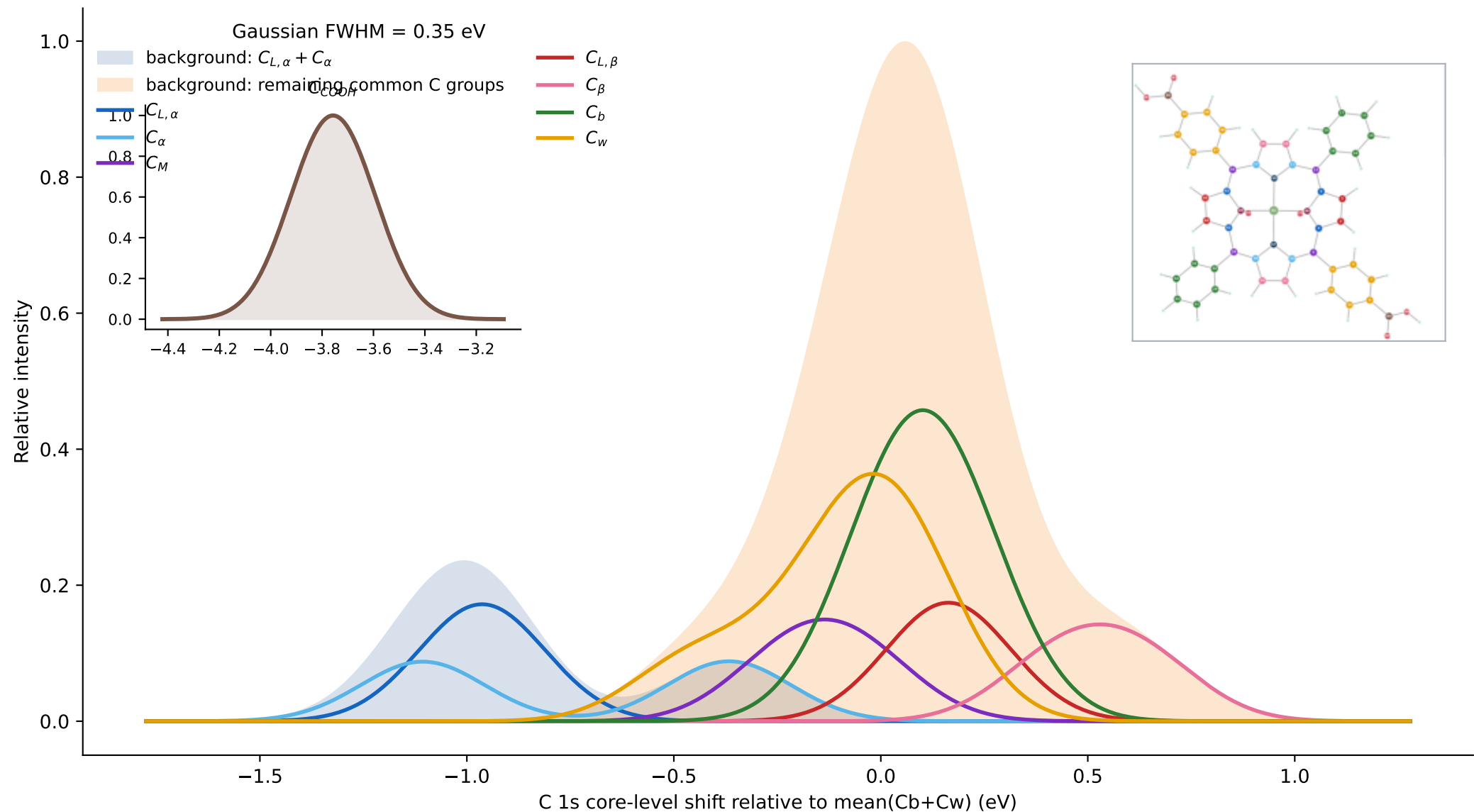
Gaussian FWHM = 0.35 eV

N_L

N



TWDCOH: C 1s initial-state core-level shifts

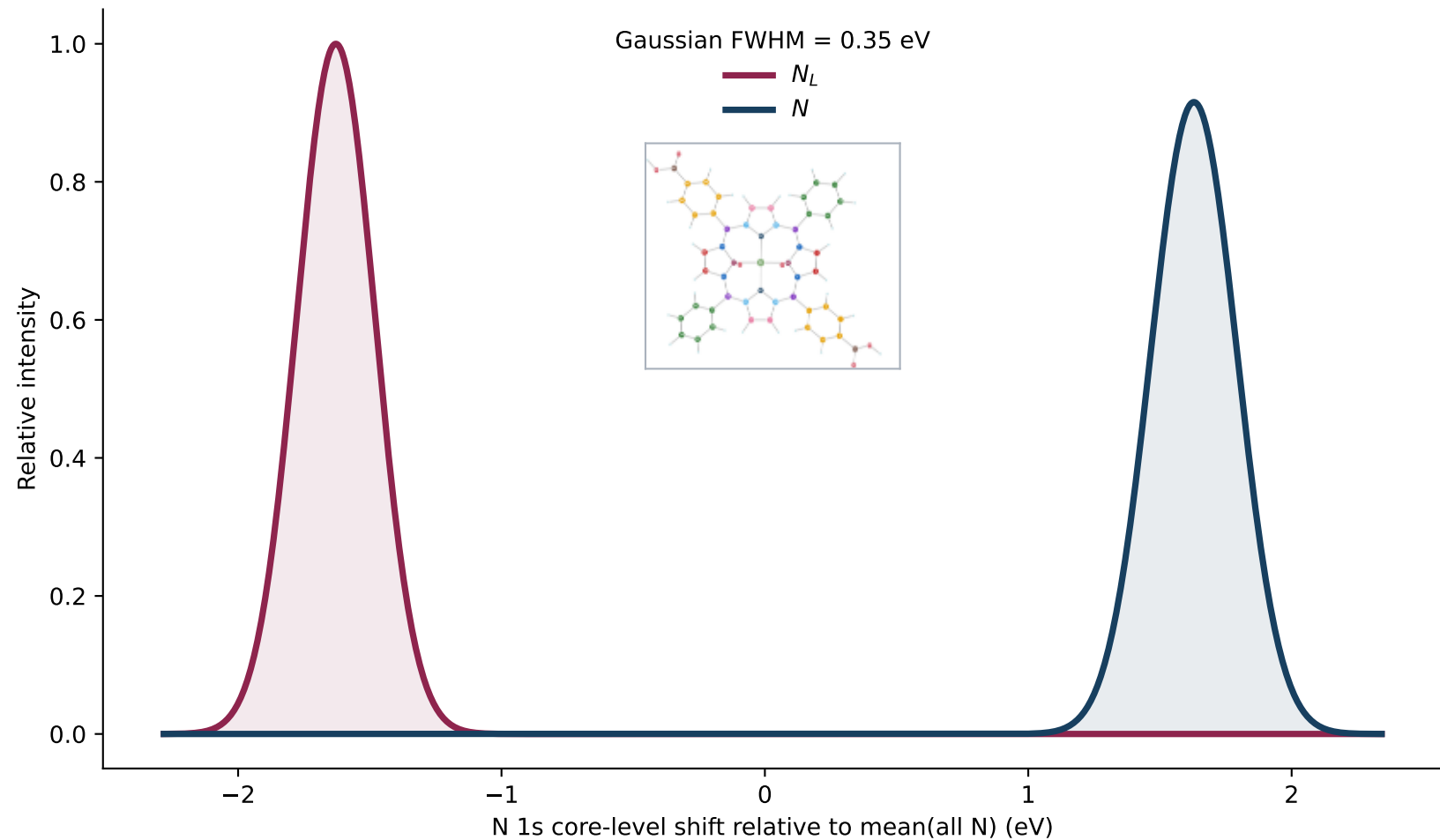
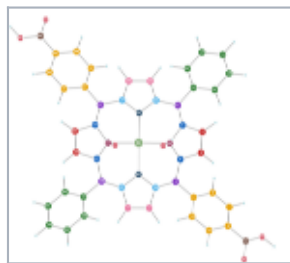


TWDCOH: N 1s initial-state core-level shifts

Gaussian FWHM = 0.35 eV

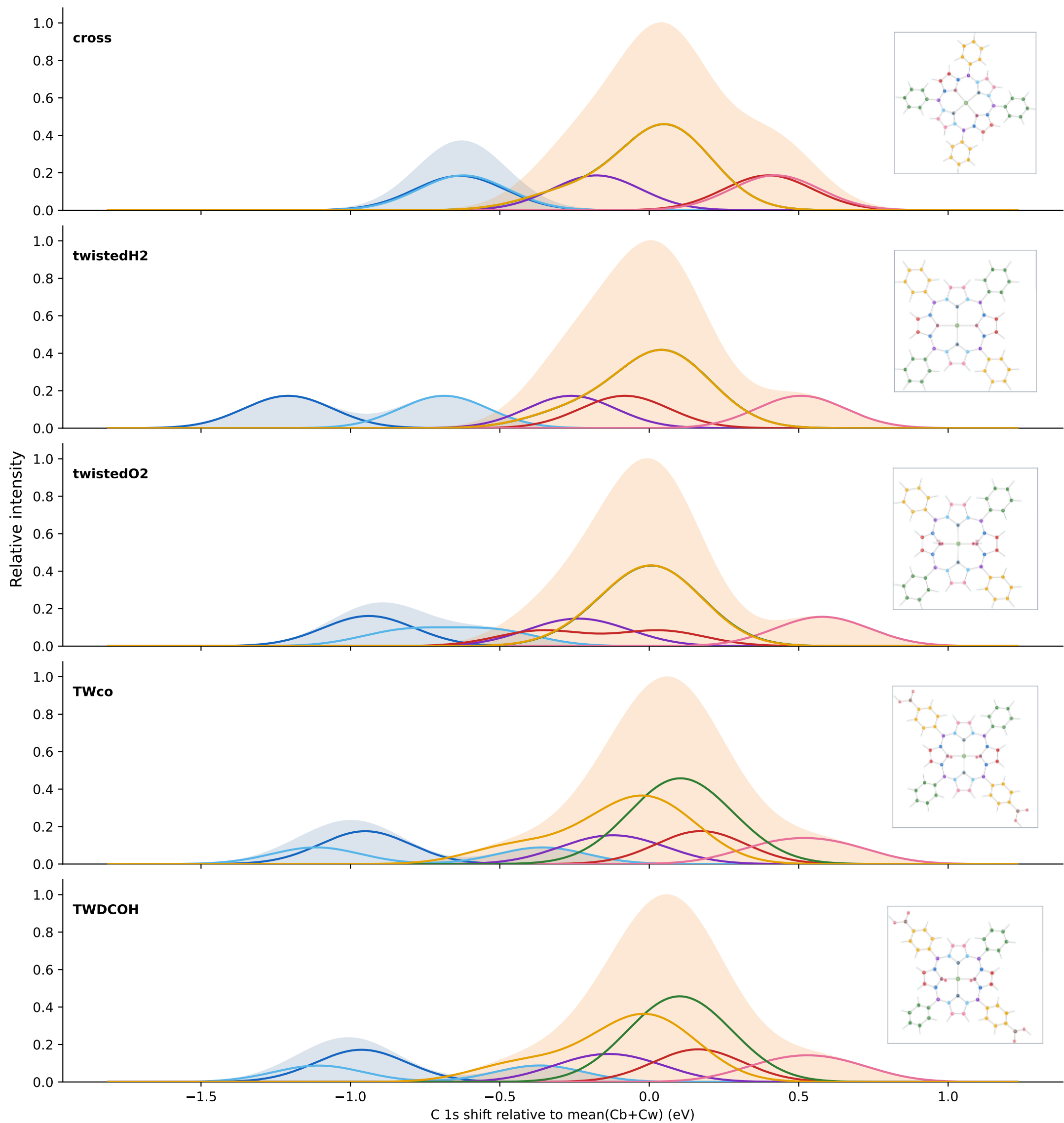
N_L

N

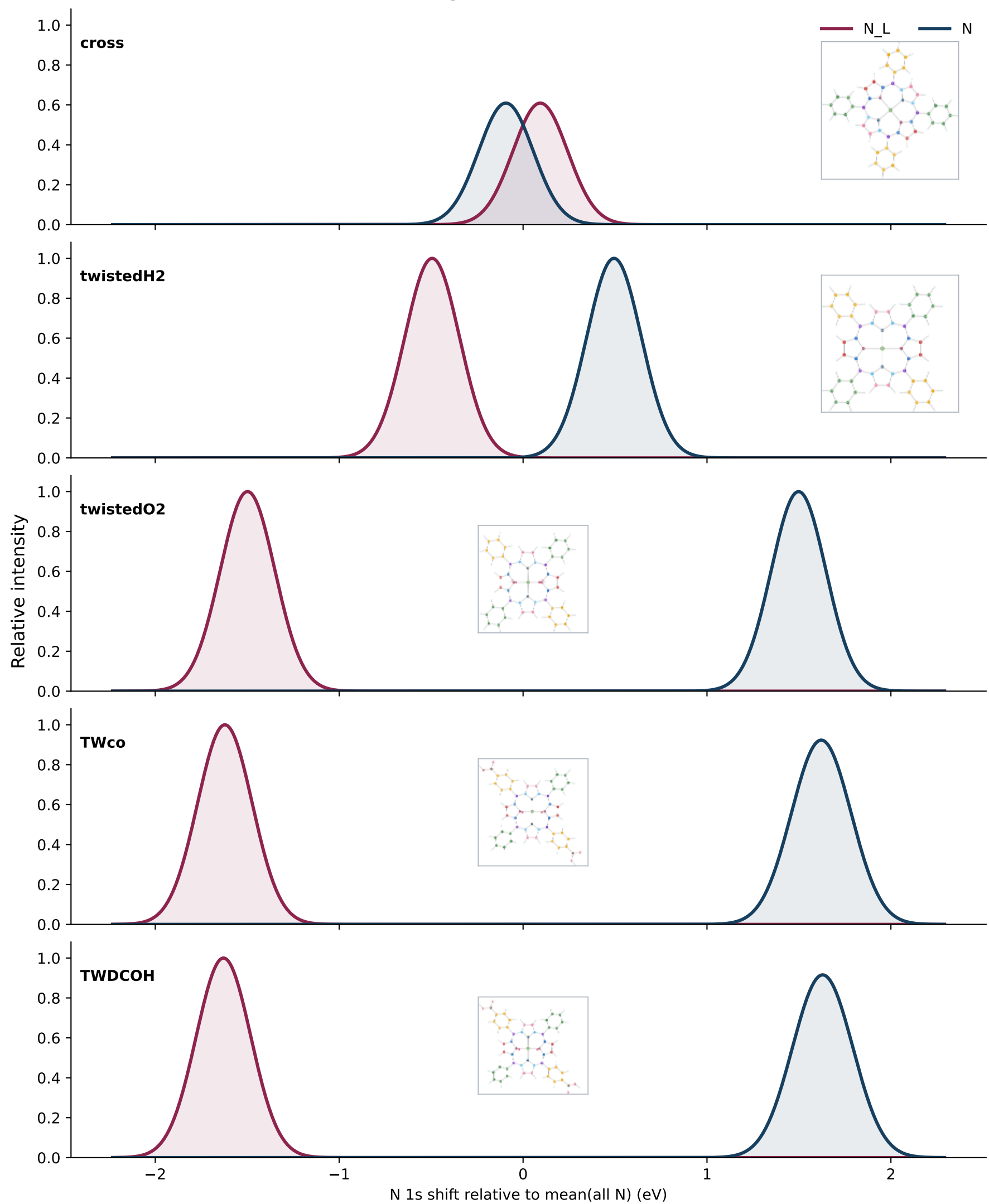


Common-carbon initial-state CLS comparison (Gaussian FWHM 0.35 eV)

$C_{L,\alpha}$
 C_α
 C_M
 $C_{L,\beta}$
 C_β
 C_b
 C_w

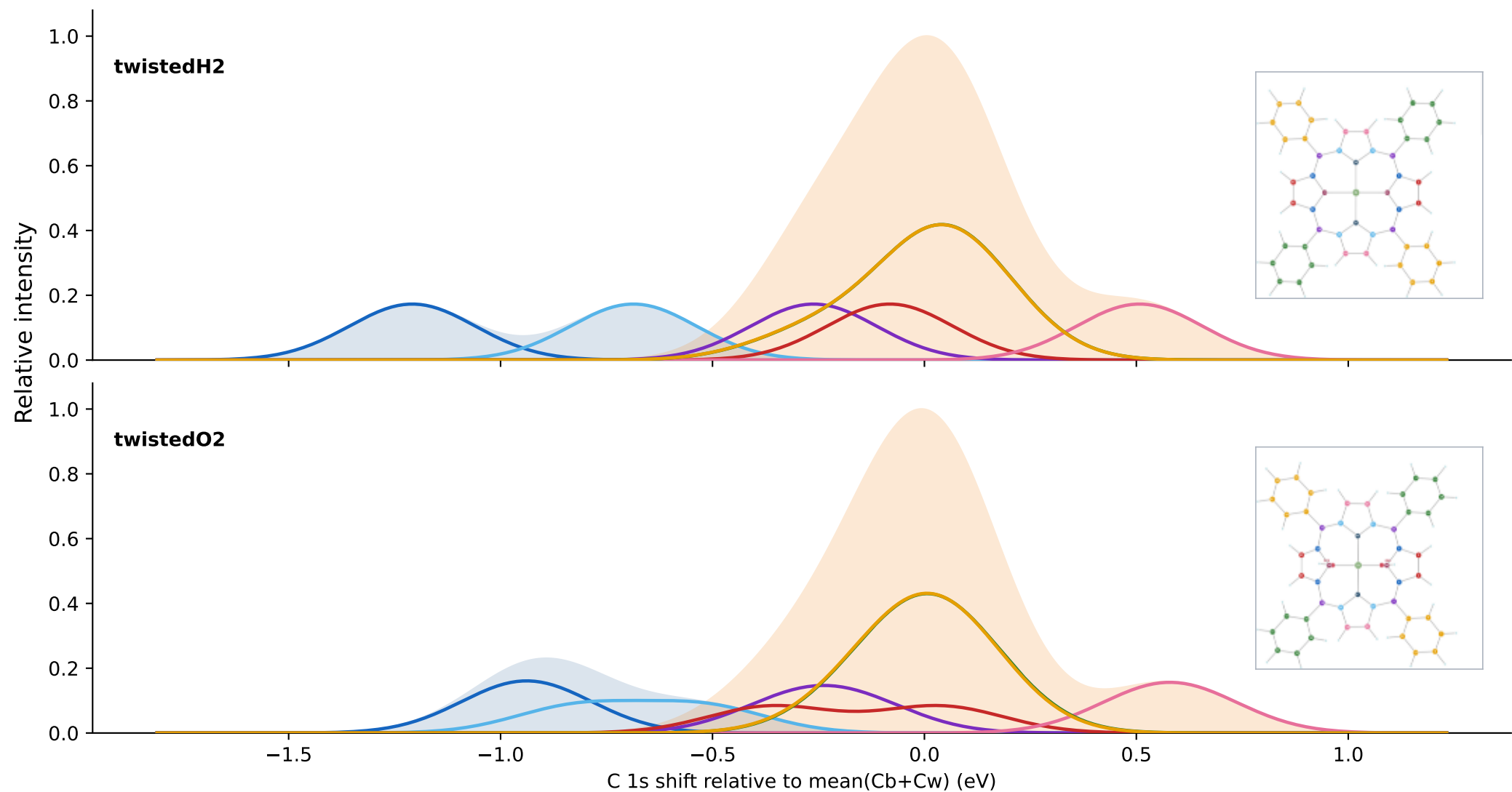


N 1s initial-state CLS comparison (Gaussian FWHM 0.35 eV)



Twisted-structure C 1s CLS comparison (Gaussian FWHM 0.35 eV)

$C_{L,\alpha} + C_\alpha$ envelope $C_{L,\alpha}$ C_M C_β C_w
other C envelope C_α $C_{L,\beta}$ C_b



Twisted-structure N 1s CLS comparison (Gaussian FWHM 0.35 eV)

